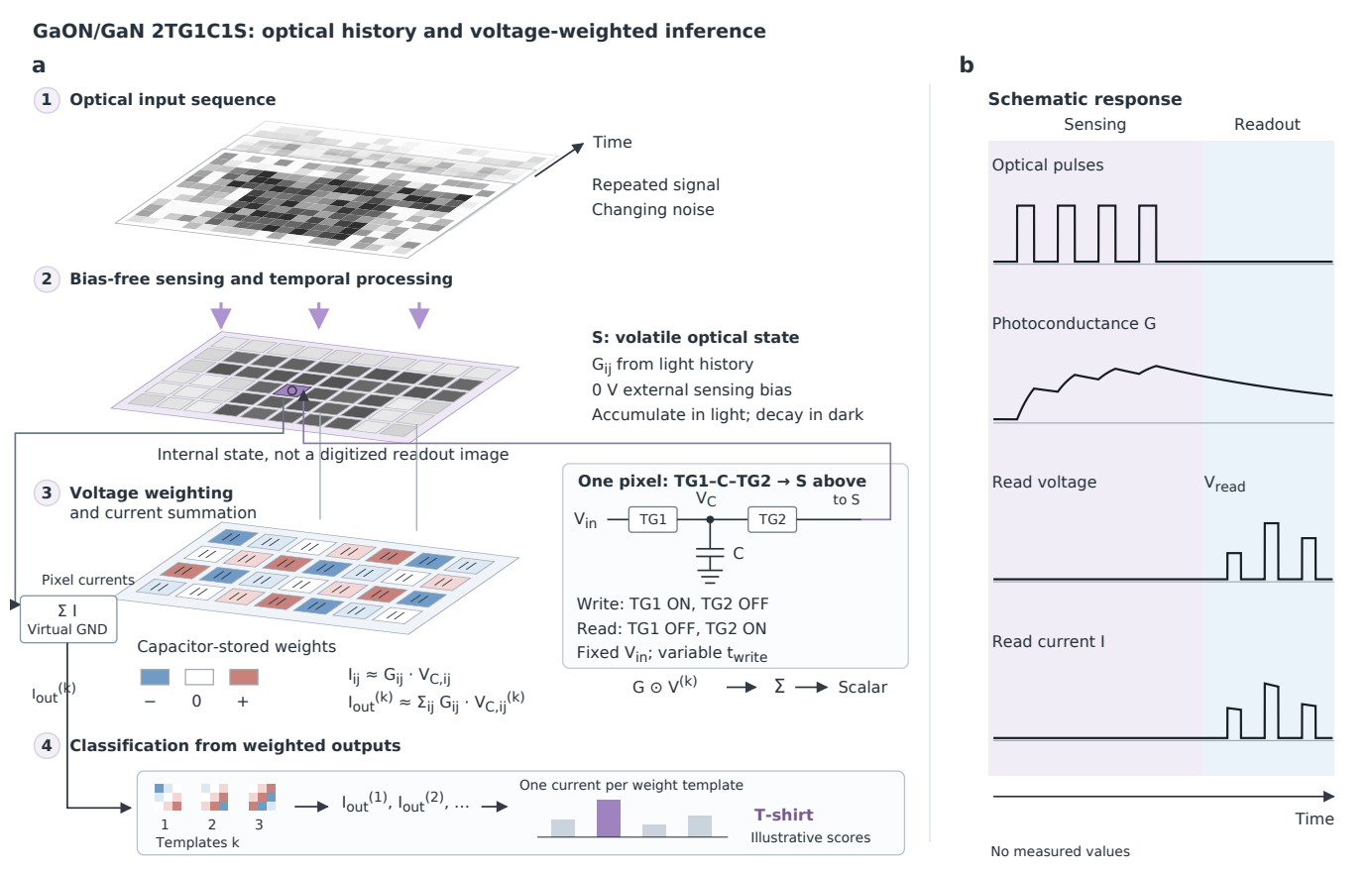

GaON/GaN 2TG1C1S: optical history and voltage-weighted inference
a
1 Optical input sequence
Time
Repeated signal
Changing noise
2 Bias-free sensing and temporal processing
S: volatile optical state
G_{ij} from light history
0 V external sensing bias
Accumulate in light; decay in dark
Internal state, not a digitized readout image
3 Voltage weighting
and current summation
Pixel currents
Σ I
Virtual GND
$I_{out}^{(k)}$
Capacitor-stored weights
− 0 +
$I_{ij} \approx G_{ij} \cdot V_{C,ij}$
$I_{out}^{(k)} \approx \Sigma_{ij} G_{ij} \cdot V_{C,ij}^{(k)}$
One pixel: TG1-C-TG2 → S above
V_C
to S
V_{in}
TG1
TG2
C
Write: TG1 ON, TG2 OFF
Read: TG1 OFF, TG2 ON
Fixed V_{in}; variable t_{write}
$G \odot V^{(k)}$ → Σ → Scalar
4 Classification from weighted outputs
1 2 3
Templates k
$I_{out}^{(1)}, I_{out}^{(2)}, \ldots$
One current per weight template
T-shirt
Illustrative scores
b
Schematic response
Sensing
Readout
Optical pulses
Photoconductance G
Read voltage
V_{read}
Read current I
Time
No measured values
Conceptual functional layers, not a physical 3D stack.